

Responsible AI

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2026-09-16





Is AI a means or obstacle in the sustainable transformation?



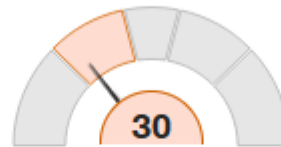
AI will take over many types of tasks and jobs

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Markets →

DOW	42,745.26	0.37% ▲
S&P 500	5,752.10	0.42% ▼
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Fear
is driving the US market

Latest Market News →

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41% of companies worldwide plan to reduce workforces by 2030 due to AI

By Olesya Dmitracova, CNN

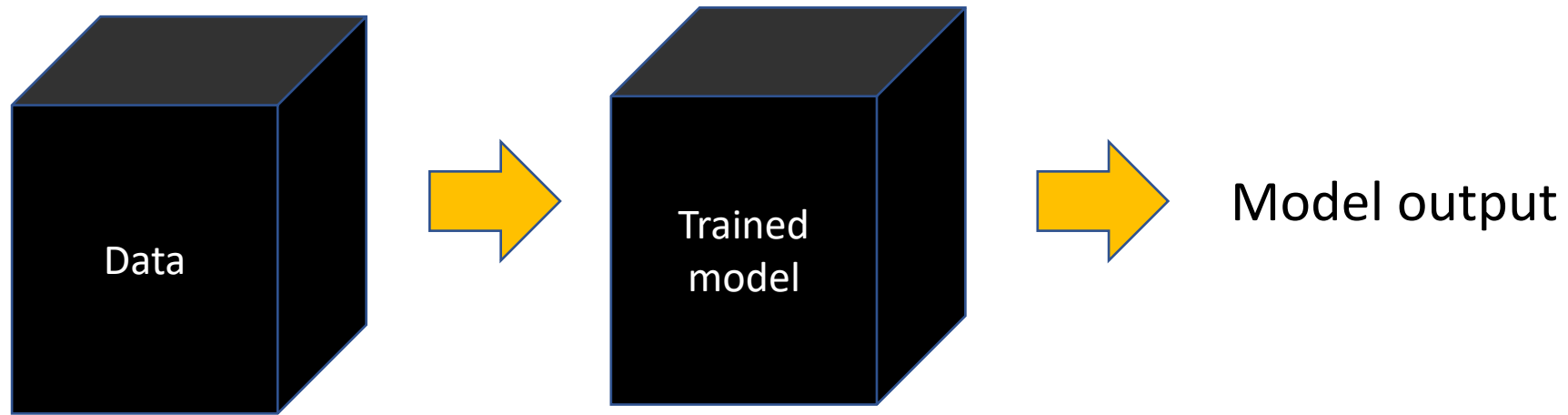
🕒 2 minute read · Published 7:57 AM EST, Wed January 8, 2025

Correct data interpretation often requires
real-world knowledge

The inspectors sent restaurant staff home
because they suspected a disease outbreak.

Deep learning models lack transparency

“Black box issue”



Explainable AI methods try to increase transparency

“Hidden” biases in the training data affect model performance

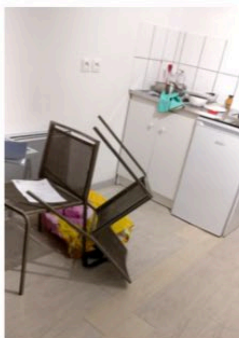
ImageNet

Chairs

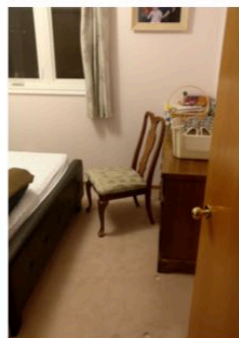


ObjectNet

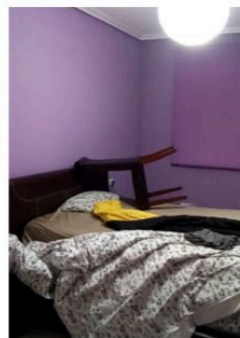
Chairs by rotation



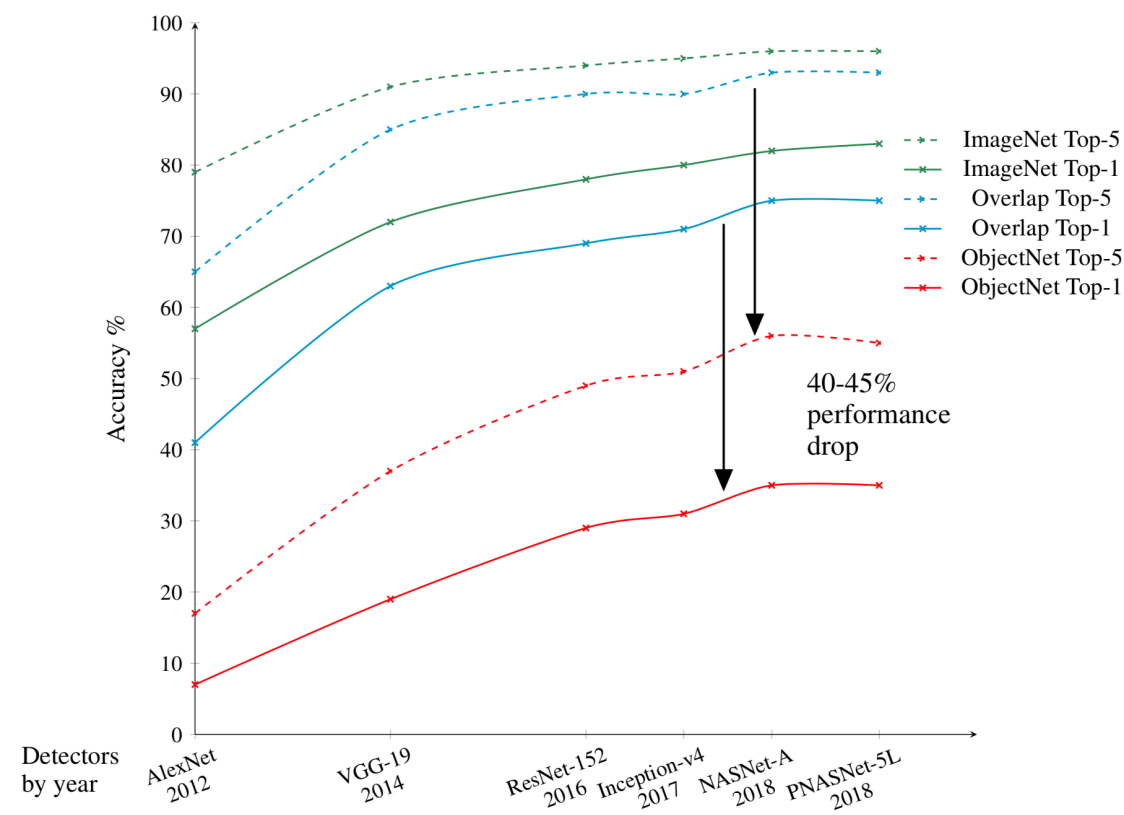
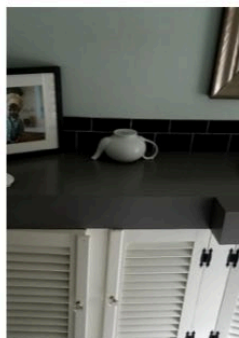
Chairs by background



Chairs by viewpoint



Teapots



Incorrect medical AI can cause real harm

AI can be prejudiced or simply wrong

Report: IBM Watson delivered ‘unsafe and inaccurate’ cancer recommendations

JULY 25, 2018 BY [FINK DENSFORD](#) — [LEAVE A COMMENT](#)



“Garbage in – Garbage out”

Data collection comes with considerable privacy risks

REGULATION

There's No Such Thing as Anonymous Data

by Scott Berinato

FEBRUARY 09, 2015

Experts warn of possible unintended use of DNA amid 23andMe bankruptcy



Washington Post technology columnist Geoffrey Fowler tells CNN's Laura Coates how DNA submitted to 23andMe may wind up being used for other purposes if the data is sold during the company's bankruptcy.

02:36 - Source: CNN

svt NYHETER Nyheter Lokalt Sport SVT Play Barn

/ INRIKES



2,7 miljoner inspelade samtal till 1177 Vårdguiden helt oskyddade på internet

Who is responsible when things
go wrong with AI?

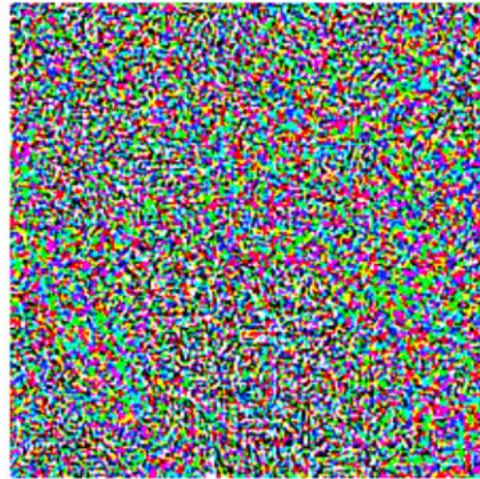
AI tools are vulnerable to malicious attacks



“panda”

57.7% confidence

+ .007 ×



noise

=



“gibbon”

99.3% confidence



AI infrastructure
is vulnerable to
malicious
attacks

Do you support open research?

What could you do with AI?

What could you do with AI if you were?

- the leader of an EU country
- the head of the WHO
- a primary care doctor in a developing nation
- a specialist in the world's best hospital
- the head of Amnesty International

TECH

Microsoft health unit is a 'multi-billion dollar business,' medical chief says

PUBLISHED THU, JUL 26 2018 • 1:36 AM EDT

TECH • APPLE

Facebook Ventures Into Medical Imaging Research

Apple's Healthcare Take Could Be \$313 Billion by 2027, Analysts Say



BY SHOBHIT SE

SCIENCE BUSINESS TECH

Google's project to 'cure death,' Calico, announces \$1.5 billion research center

67

Calico will work with big pharma to create and market new drugs for diseases like cancer and Alzheimer's

By [Ben Popper](#) | Sep 3, 2014, 2:33pm EDT

Via [Re/code](#) | Source [Calico Labs](#)



BUS



Column: Alexa may be key to Amazon's looming domination of the healthcare market

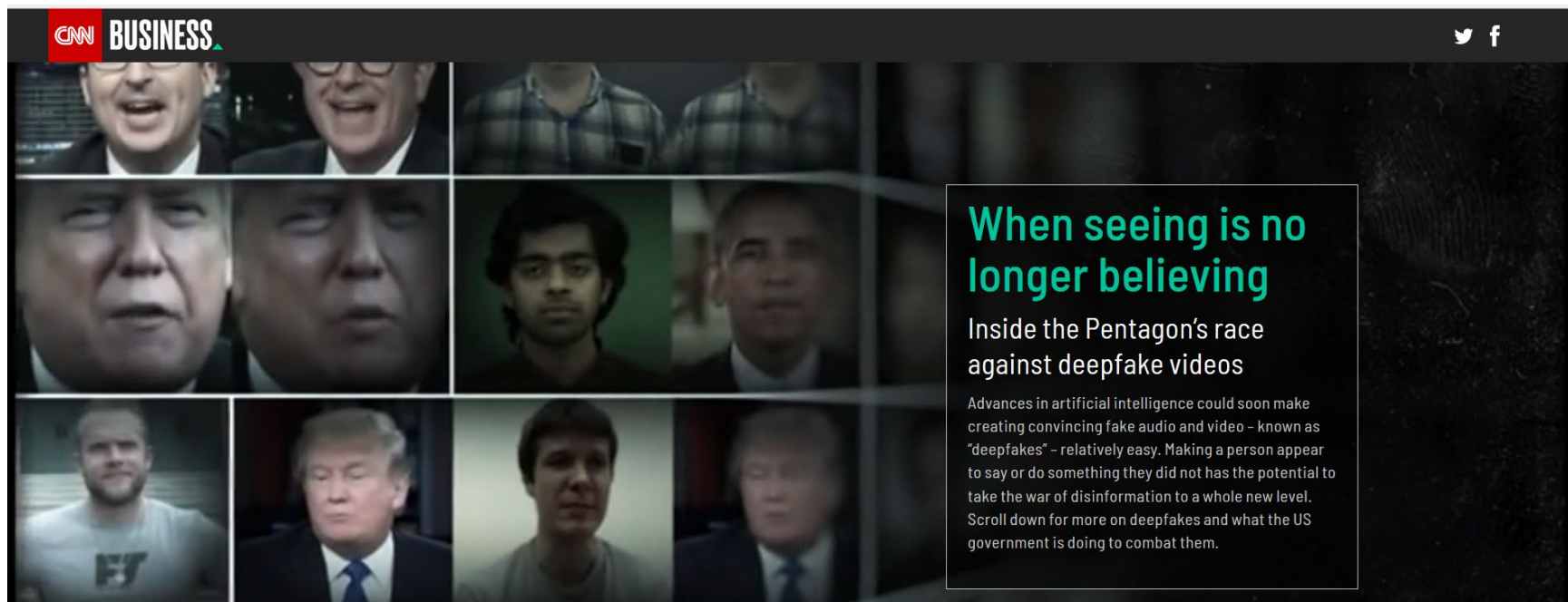
What could you do with AI if you were?

- the leader of an EU country
- the head of the WHO
- a primary care doctor in a developing nation
- a specialist in the world's best hospital
- the head of Amnesty International
- the leader of a global tech company

OpenAI has published the text-generating AI it said was too dangerous to share

The lab says it's seen 'no strong evidence of misuse so far'

By James Vincent | Nov 7, 2019, 7:24am EST





The New York Times



China Uses DNA to Track Its People, With the Help of American Expertise

The Chinese authorities turned to a Massachusetts company and a prominent Yale researcher as they built an enormous system of surveillance and control.

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CONTINUE

What could you do with AI if you were?

- the leader of an EU country
- the head of the WHO
- a primary care doctor in a developing nation
- a specialist in the world's best hospital
- the head of Amnesty International
- the leader of a global tech company
- a terrorist
- a dictator
- a white supremacist
- a criminal

Do you think AI application in
your research domain fall under
dual use?

Should we have open research in
the field of AI?

AI **for** sustainability



Sustainability **of** AI

Energy



Researcher

Supercomputer

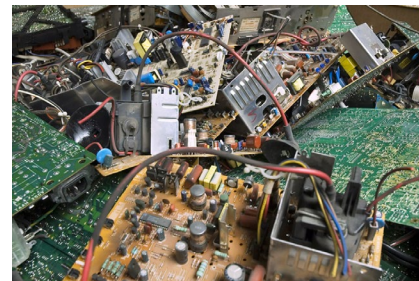
Transport

Production

Raw materials



Waste and pollution



Exponential growth of parameters in notable AI systems

Our World
in Data

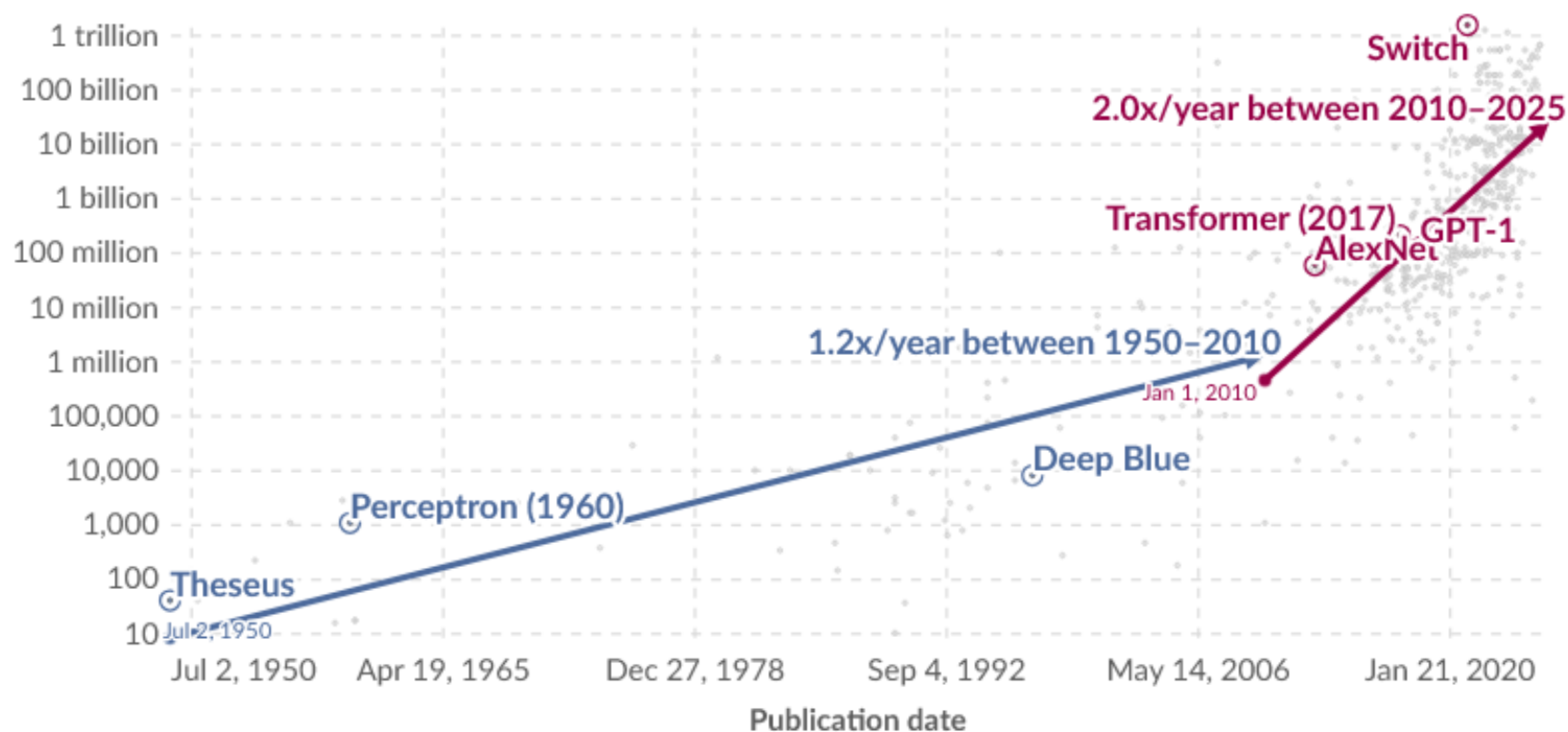
Parameters are variables in an AI system whose values are adjusted during training to establish how input data gets transformed into the desired output; for example, the connection weights in an artificial neural network.

Table Chart

Select systems

Settings

Number of parameters



Jul 2, 1950



Mar 6, 2025

State-of-the-art models cost > 100 million dollars to train
→ AI dominated by a few large companies (mostly US/China)

Estimated training cost of select AI models, 2016–23

Source: Epoch, 2023 | Chart: 2024 AI Index report

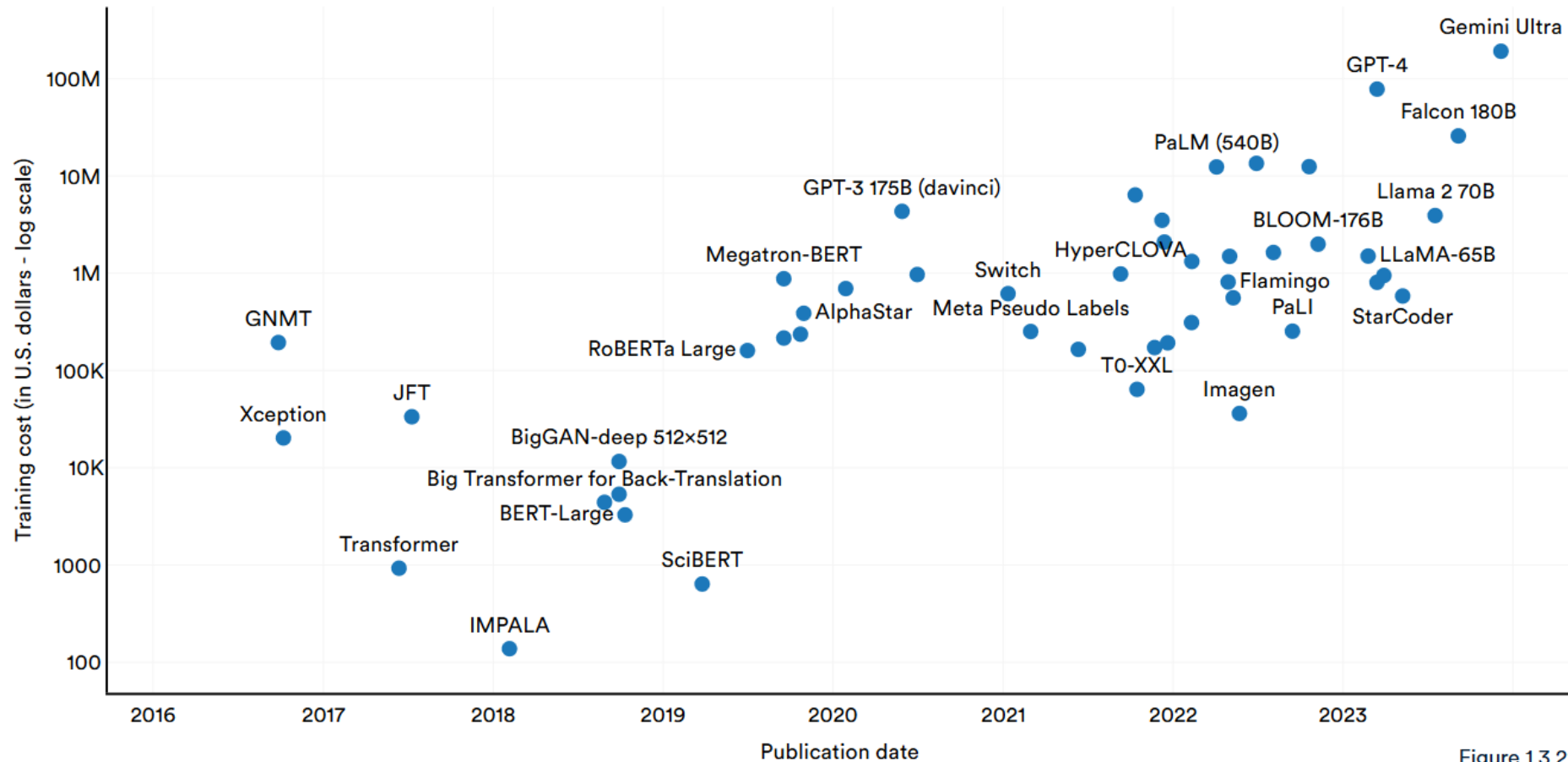


Figure 1.3.22

Can we trust AI models?
Can we control AI?

'Godfather of AI' shortens odds of the technology wiping out humanity over next 30 years

Geoffrey Hinton says there is 10% to 20% chance AI will lead to human extinction in three decades, as change moves fast

- 'We need dramatic changes': is societal collapse inevitable?



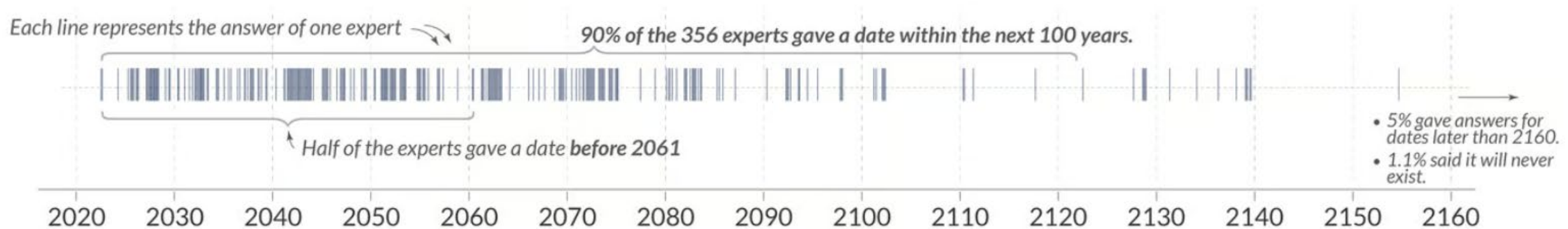
Will we have Artificial General Intelligence soon?

When will there be a 50% chance that Human-level Artificial Intelligence exists?



Timelines of **356 AI experts**, surveyed in **2022** by Katja Grace and colleagues.

The experts were asked when unaided machines will be able to accomplish every task better and more cheaply than human workers.



Data from Zach Stein-Perlman, Benjamin Weinstein-Raun, Katja Grace – 2022 Expert Survey on Progress in AI.

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EU Ethics guidelines for trustworthy AI

- (1) lawful - respecting all applicable laws and regulations (e.g. EU AI act)
- (2) ethical - respecting ethical principles and values
- (3) robust - both from a technical perspective while taking into account its social environment

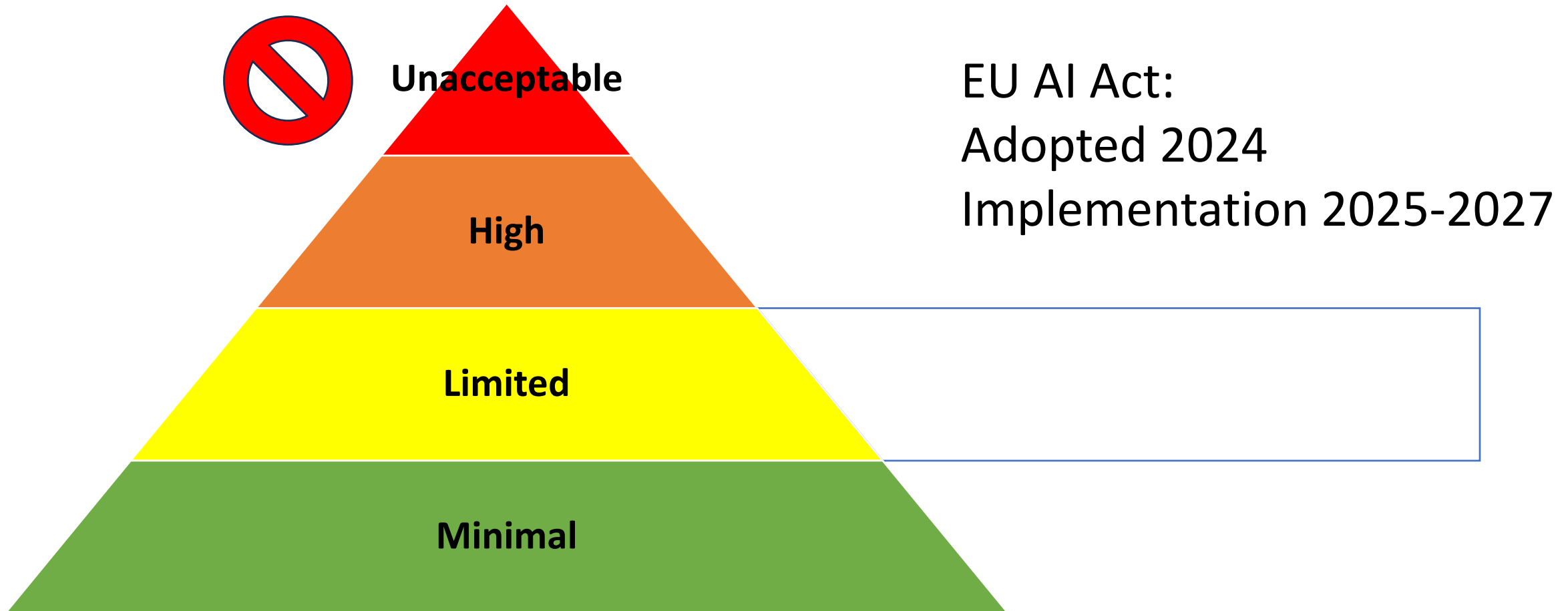
<https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai>

7 requirements for ethical AI

- Human agency and oversight
- Technical Robustness and safety
- Privacy and data governance
- Transparency
- Diversity, non-discrimination and fairness
- Societal and environmental well-being
- Accountability

<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>

AI regulation is struggling to keep up with rapid technology development



There are many ethical, legal, environmental and societal issues for each case

- Ownership of the training data, AI tools and AI output
- Sources and risks of bias
- Transparency regarding development and output
- Societal benefits and negative impact (e.g. on the job market)
- Environmental benefits and negative impact
- Accessibility (e.g. for non-English speakers, low-income countries, people with handicaps)
- Possibilities for abuse by malicious actors
- Responsibility for correctness and safety
- Possibilities for oversight by public authorities and by the end-user
- Consequences of forbidding the development and usage of this type of AI in the EU

Successful AI
projects require
successful
interdisciplinary
collaboration

- Life science
- AI/data science
- Data management/engineering
- MLOps/AI engineering
- IT infrastructure/security
- Software development/engineering
- Project management
- Regulatory/quality
- Ethics/sustainability
- Business operations/intelligence

How can we all contribute to a
ethical and sustainable use of AI?